

Cryptography

PRF+OTP and Modes of Operation

Arkady Yerukhimovich

September 21, 2026

- Homework 3 is due next Monday (September 28) at 12:45pm.
- Research project ideas and deliverables document out on Piazza.
Contact me to discuss your topics. Approved project topics are due no later than October 26.
- Office hours this Thursday are cancelled. Ping me on Piazza or email to schedule a meeting if you need help.

- 1 PRF Review and Quiz
- 2 Security of PRF+OTP
- 3 Modes of Operation

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Quiz

Break

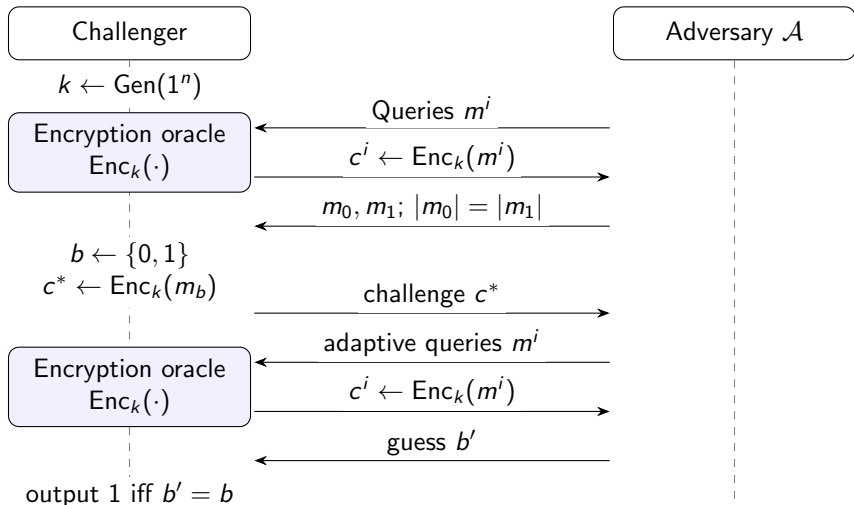
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Where We Stopped Last Time

Goal: Build a CPA-secure encryption scheme from a PRF

- Secure against a party who gets to see encryptions of messages of their choice (chosen-plaintext attack)
- Can encrypt many messages

The CPA Security Game



CPA-Secure Encryption from a PRF

PRF+OTP Encryption (Π)

- $\text{Gen}(1^n)$: $k \leftarrow \{0, 1\}^n$
- $\text{Enc}(k, m)$: Choose $r \leftarrow \{0, 1\}^n$, output $c = (r, F_k(r) \oplus m)$
- $\text{Dec}(k, c)$: Parse c as (r, c') , compute $m = F_k(r) \oplus c'$

Theorem

If F is a secure PRF, then PRF+OTP is CPA-secure

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- ② Use a probabilistic argument to prove that scheme is unconditionally secure when using a random function f .
 - Random function is essentially a collection of 2^n OTPs
 - Proof is similar to proof of OTP, but need to account for probability of collision in r

Security of PRF+OTP: Step 1

Define the following encryption scheme $\tilde{\Pi}$:

$\tilde{\Pi}$ Encryption Scheme

- $\widetilde{\text{Gen}}(1^n)$: $f \leftarrow \mathcal{F}_n$ (the set of functions $\{0, 1\}^n \rightarrow \{0, 1\}^n$)
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- Observe that this is exactly PRF+OTP with F_k replaced by f
 - This encryption is not efficient as we cannot evaluate a random function
 - But, it is useful as a “thought experiment” in the proof as it gives us a target for security

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Lemma: For any PPT $\mathcal{A}$

$$\left| \Pr[\text{PrivK}_{\mathcal{A}, \Pi}^{cpa}(n) = 1] - \Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{cpa}(n) = 1] \right| \leq \text{negl}(n)$$

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- Use this to construct $\mathcal{A}_r$ that breaks PRF security of F_k

The Two Adversaries

$PRF_{D,F}(n)$

- The challenger chooses $b \leftarrow \{0, 1\}$.
If $b = 0$, he chooses $f \leftarrow \mathcal{F}_n$ and gives $\mathcal{D}$ an oracle $\mathcal{O} = f$.
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- With access to oracle $\mathcal{O}$, the distinguisher $\mathcal{D}$ outputs a bit b'
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 - We care about the *difference* in $\mathcal{A}_c$'s WIN probability

Constructing $\mathcal{A}_r^{\mathcal{O}}$: Intuition

- $\mathcal{A}_r$ needs to use $\mathcal{A}_c$ to win PRF game

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- $\mathcal{A}_r$ acts as the challenger for $\mathcal{A}_c$ in CPA-security game
 - $\mathcal{A}_r$ must answer $\mathcal{A}_c$'s $\text{Enc}(\cdot)$ queries (i.e., simulate the Enc oracle)
 - $\mathcal{A}_r$ must produce the challenge ciphertext c

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 - $\mathcal{A}_r$ must answer $\mathcal{A}_c$'s $\text{Enc}(\cdot)$ queries (i.e., simulate the Enc oracle)
 - $\mathcal{A}_r$ must produce the challenge ciphertext c
- If $\mathcal{A}_c$ WINS, $\mathcal{A}_r$ must use that to win the game against his challenger

① Pre-Challenge

Run $\mathcal{A}_c(1^n)$ and when $\mathcal{A}_c$ asks $\text{Enc}(m)$ query

- Choose $r \leftarrow \{0, 1\}^n$, query $y = \mathcal{O}(r)$, return $c = (r, y \oplus m)$ to $\mathcal{A}_c$

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When $\mathcal{A}_c$ outputs (m_0, m_1)

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3 Post-Challenge

Continue answering Enc queries until $\mathcal{A}_c$ outputs guess b'

- Output 1 (“PRF”) if $b = b'$, and 0 otherwise.

Observation

- If $\mathcal{O}$ is f , then $\mathcal{A}_r$ is simulating $\tilde{\Pi}$
- If $\mathcal{O}$ is F_k , then $\mathcal{A}_r$ is simulating Π

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- We assumed that $\mathcal{A}_c$ breaks the lemma – i.e. has different success probability vs. Π and $\tilde{\Pi}$

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- That is, $\mathcal{A}_r$ is able to distinguish between $F_k(\cdot)$ and $f(\cdot)$. But, we know that F_k is a PRF.

Contradiction!

To prove security from a PRF, we often do the following:

- ✓ Consider the scheme where F_k is replaced by a random function f
 - Show by reduction to security of PRF, that $\mathcal{A}$ can't tell we made this change.
 - So, $\mathcal{A}$'s success probability must be (essentially) the same in this and original variant.
- ② Use a probabilistic argument to prove that scheme is unconditionally secure when using a random function f .
 - Random function is essentially a collection of 2^n OTPs
 - Proof is similar to proof of OTP, but need to account for probability of collision in r

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Lemma

For any $\mathcal{A}$ making at most $q(n)$ queries to $\text{Enc}(\cdot)$

$$\Pr[\text{PrivK}_{\mathcal{A}, \tilde{\Pi}}^{\text{cpa}}(n) = 1] \leq 1/2 + \frac{q(n)}{2^n}$$

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 - $\mathcal{A}$ learns value of $f(r^*)$ (he sees $c = (r^*, c')$, computes $f(r^*) = c' \oplus m$)
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- So,

$$\Pr[r^* \in \{r_1, \dots, r_{q(n)}\}] \leq \sum_{i=1}^{q(n)} \Pr[r^* = r_i] = \frac{q(n)}{2^n} \leq \text{negl}(n)$$

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Finishing Proof of CPA-security of PRF+OTP

- ✓ Consider the scheme where F_k is replaced by a random function f
 - We showed that any PPT $\mathcal{A}$ has only a $\text{negl}(n)$ advantage in distinguishing the two games
- ✓ Use a probabilistic argument to prove that scheme is unconditionally secure when using a random function f .
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Combining these two statements, we get that for any PPT $\mathcal{A}$,

$$\Pr[\text{PrivK}_{\mathcal{A}, \text{PRF+OTP}}^{\text{cpa}}(n) = 1] \leq 1/2 + \frac{q(n)}{2^n} + \text{negl}(n)$$

- 1 PRF Review and Quiz
- 2 Security of PRF+OTP
- 3 Modes of Operation**

Encrypting Long Messages

CPA-secure encryption allows us to

- Encrypt many messages using the same key k

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The Question

How do we encrypt long messages without this 2X increase in size

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- Modes of operation study how to encrypt many block messages without blow-up in size

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- Modes of operation study how to encrypt many block messages without blow-up in size
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- Can think of them as other ways to turn PRF into CPA-secure encryption

Modes of Operation: Important Properties

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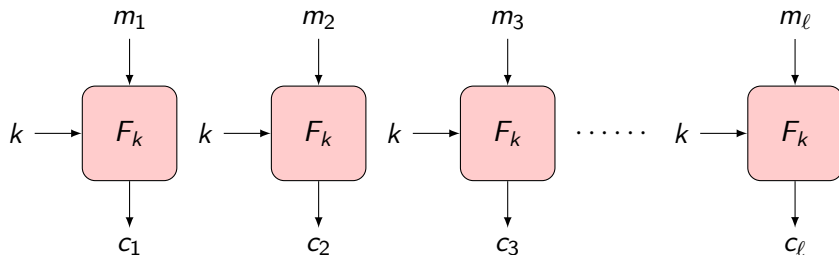
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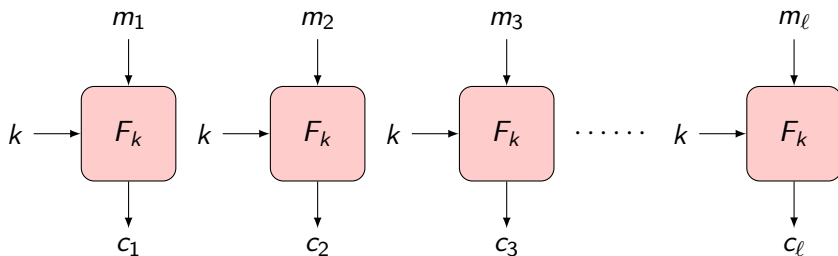
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- Security - want at least CPA-security

Electronic Code Book (ECB) Mode

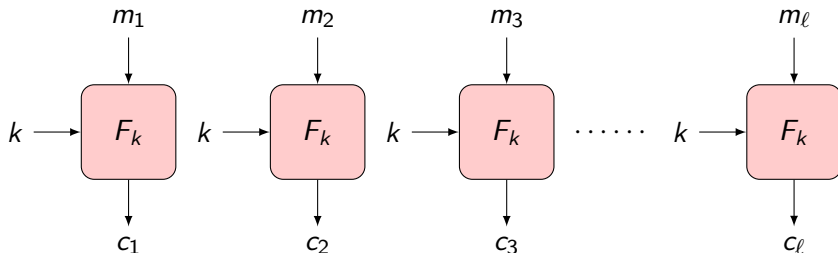


Electronic Code Book (ECB) Mode



- Ciphertext length: $|c| = |m|$
- Pre-computation: N/A
- Parallelism: Can encrypt/decrypt blocks in parallel
- PRF Inverse: Need to compute inverse to decrypt
- Security: ??

Electronic Code Book (ECB) Mode



- Ciphertext length: $|c| = |m|$
- Pre-computation: N/A
- Parallelism: Can encrypt/decrypt blocks in parallel
- PRF Inverse: Need to compute inverse to decrypt
- Security: NOT SECURE
 - ECB Mode is deterministic
 - Can tell if two blocks are the same

Electronic Code Book (ECB) Mode: How bad is it?

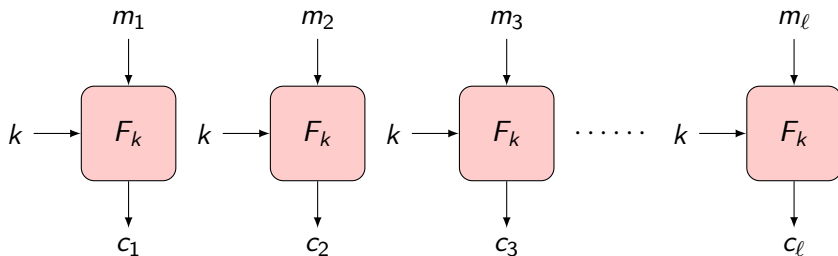


Figure: Original Image

Electronic Code Book (ECB) Mode: How bad is it?

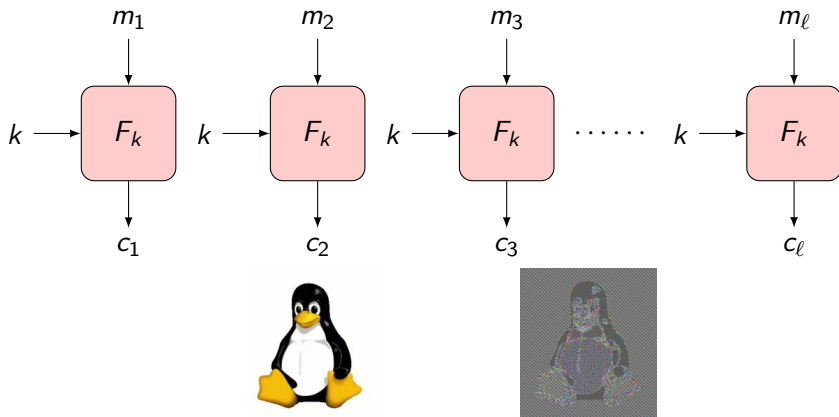


Figure: Original Image Figure: ECB-encrypted

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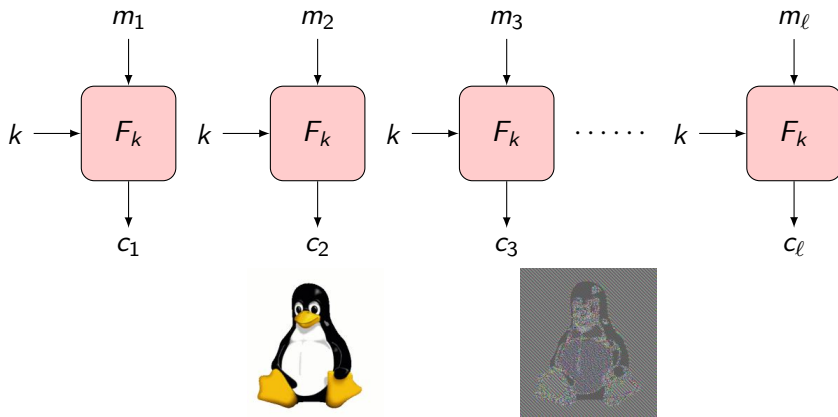
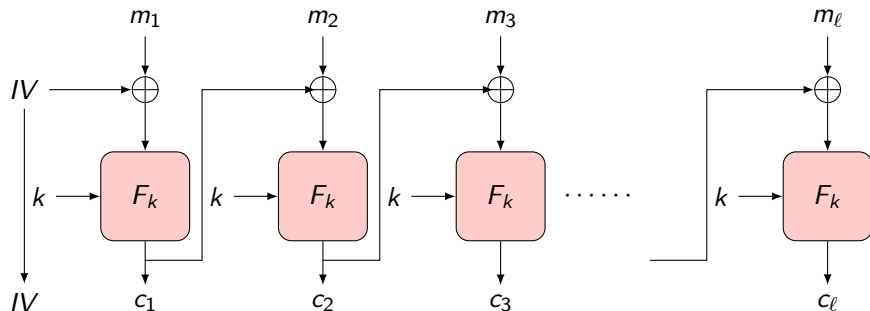


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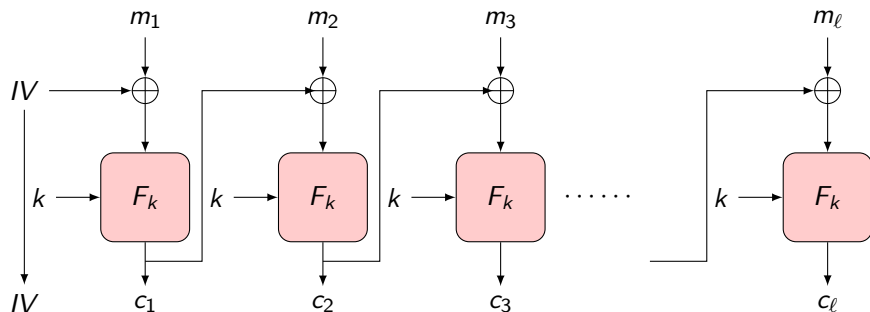
Warning

Never use ECB mode

Cipher Block Chaining (CBC) Mode



Cipher Block Chaining (CBC) Mode



- Ciphertext length: $|c| = |m| + 1$ blocks
- Pre-computation: N/A
- Parallelism: encryption sequential; decryption parallel
- PRF inverse: needed for decryption
- Security: CPA-secure with a random IV